

Cloud Computing Career Roadmap

A complete, self-paced path from beginner to job-ready — using 100% free resources

How to use this roadmap

This roadmap has 6 phases. It focuses on AWS as the primary cloud (largest market share) while also pointing you to Azure/GCP equivalents, since core concepts transfer across all three. Expect 8–12 months of consistent part-time study (5–10 hrs/week) to become job-ready for an entry-level cloud role.

Phase 1 — Foundations (Weeks 1–6)

Goal: Build the fundamentals every cloud role assumes you already know.

Topics to cover

- Networking basics: IP addressing, subnets, DNS, firewalls, load balancers
- Linux fundamentals: file system, permissions, package managers, shell scripting
- Basic programming: Python or Bash for automation
- Version control: Git & GitHub fundamentals

Free resources

Networking & Linux

- Professor Messer — free Network+ course ([professormesser.com](https://www.professormesser.com))
- Linux Journey — free interactive course (linuxjourney.com)

Python & Git

- freeCodeCamp — 'Python for Everybody' (YouTube, full course)
- Git & GitHub Crash Course — freeCodeCamp (YouTube)

Phase 2 — Core Cloud Concepts (Weeks 7–12)

Goal: Understand what cloud computing actually is and how providers structure services.

Topics to cover

- IaaS vs PaaS vs SaaS, public/private/hybrid cloud
- Core services: compute, storage, networking, databases
- Cloud economics: pricing models, the free tier, shared responsibility model
- Regions, availability zones, and high availability concepts

Free resources

Structured Intro Courses

- AWS Cloud Practitioner Essentials — free official AWS course (aws.amazon.com/training)
- Microsoft Learn — Azure Fundamentals (AZ-900) learning path, 100% free (learn.microsoft.com)
- Google Cloud Skills Boost — free introductory quests (cloudskillsboost.google, free labs monthly)

Phase 3 — Hands-On with a Cloud Provider (Weeks 13–22)

Goal: Get real practical experience deploying and managing cloud resources.

Topics to cover (using AWS as primary)

- EC2 (virtual machines), S3 (storage), VPC (networking)
- IAM: users, roles, policies, least-privilege access
- RDS/DynamoDB (databases), Lambda (serverless basics)
- CloudWatch: monitoring & logging basics

Free resources

Free Hands-On Labs

- AWS Free Tier — 12 months free access to core services (aws.amazon.com/free)
- AWS Skill Builder — free digital courses & labs (skillbuilder.aws)
- Microsoft Learn Sandbox — free hands-on Azure labs, no credit card needed
- Google Cloud Skills Boost — free monthly credits for hands-on labs
- Qwiklabs free monthly quests (via Google Cloud Skills Boost)

Guided Project-Based Learning

- freeCodeCamp — 'AWS Certified Cloud Practitioner' full free course (YouTube, 14+ hrs)
- KodeKloud free courses — hands-on browser-based labs (kodekloud.com, free tier)

Phase 4 — DevOps & Automation Skills (Weeks 23–32)

Goal: Learn the tools that make you valuable beyond basic cloud console usage.

Topics to cover

- Infrastructure as Code: Terraform basics
- Containers: Docker fundamentals, then Kubernetes basics
- CI/CD pipelines: GitHub Actions or Jenkins basics
- Configuration management: Ansible basics (optional but valuable)

Free resources

IaC & Containers

- HashiCorp Learn — free official Terraform tutorials (developer.hashicorp.com/terraform/tutorials)
- Docker's official 'Get Started' free guide (docs.docker.com/get-started)
- Kubernetes official free tutorial (kubernetes.io/docs/tutorials)
- freeCodeCamp — 'Docker and Kubernetes Full Course' (YouTube)

CI/CD

- GitHub Actions official free docs & quickstart (docs.github.com/actions)
- TechWorld with Nana — free DevOps roadmap videos (YouTube)

Phase 5 — Free & Low-Cost Certifications (Weeks 33–38)

Certifications prove your skills to employers. Study fully free, pay only for the exam if desired:

Recommended Certification Path

- AWS Certified Cloud Practitioner — study 100% free via AWS Skill Builder + freeCodeCamp course
- Microsoft Azure Fundamentals (AZ-900) — free study path on Microsoft Learn
- Google Associate Cloud Engineer — free study path on Cloud Skills Boost

Note: Exam vouchers typically cost \$99–\$150. Study is 100% free; only the certification exam itself has a cost. Look for free voucher promotions in official provider communities.

Phase 6 — Specialize & Build Portfolio (Weeks 39–48)

Choose a specialization

- **Cloud Architecture:** Designing scalable, resilient systems
- **DevOps/SRE:** CI/CD, automation, reliability engineering
- **Cloud Security:** Pairs directly with the Cybersecurity roadmap
- **Data Engineering on Cloud:** Data pipelines, warehousing (BigQuery, Redshift)

Build your portfolio

- Deploy a full 3-tier web app on AWS free tier and document the architecture
- Automate deployment with Terraform + GitHub Actions and publish the code on GitHub
- Write blog posts on your learning journey (great for interviews and visibility)

Free resources

Portfolio & Job Prep

- GitHub Pages — free project hosting & blog (pages.github.com)
- Forage.com — free virtual job simulations from real cloud/tech companies
- [r/aws](#), [r/devops](#), [r/googlecloud](#) — free community Q&A; on Reddit

Suggested weekly rhythm: 30% theory, 50% hands-on labs/projects, 20% documenting/writing. Building real, deployed projects is what actually gets you hired — prioritize labs over passive video watching.